Similarly, by a suitable change in motion you can create a force that is essentially identical to gravity. For example, imagine that Barney joins astronauts floating weightless in their space capsule, with the bathroom scale still stuck to his feet and still reading zero. If the capsule should fire up its boosters and accelerate, things will change significantly. Barney will feel pressed to the capsule's floor, just as you feel pressed to the floor of an upward accelerating elevator. And since Barney's feet are now pressing against the scale, its reading is no longer zero. If the captain fires the boosters with just the right oomph, the reading on the scale will agree precisely with what Barney saw in the bathroom. Through appropriate acceleration, Barney is now experiencing a force that is indistinguishable from gravity.

The same is true of other kinds of accelerated motion. Should Barney join Homer in the outer space bucket, and, as the bucket spins, stand at a right angle to Homer—feet and scale against the inner bucket wall—the scale will register a nonzero reading since his feet will press against it. If the bucket spins at just the right rate, the scale will give the same reading Barney found earlier in the bathroom: the acceleration of the spinning bucket can also simulate earth's gravity.

All this led Einstein to conclude that the force one feels from gravity and the force one feels from acceleration are the same. They are equivalent. Einstein called this the *principle of equivalence.*

Take a look at what it means. Right now you feel gravity's influence. If you are standing, your feet feel the floor supporting your weight. If you are sitting, you feel the support somewhere else. And unless you are reading in a plane or a car, you probably also think that you are stationary—that you are not accelerating or even moving at all. But according to Einstein you actually are accelerating. Since you're sitting still this sounds a little silly, but don't forget to ask the usual question: Accelerating according to what benchmark? Accelerating from whose viewpoint?

With special relativity, Einstein proclaimed that absolute spacetime provides the benchmark, but special relativity does not take account of gravity. Then, through the equivalence principle, Einstein supplied a more robust benchmark that does include the effects of gravity. And this